Supplementary Information for: Decoder decay and its correction over years in intracortical brain–computer interfaces

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Abstract

This document contains Supplementary Notes 1–4 and Supplementary Tables 1–14. Every table is generated from the released result files by `scripts/make_supp_tables.py`, so its values match the code repository.

Supplementary Note 1: Label-free monitoring statistics

Before the main analyses, we asked whether label-free statistics computed on a later day can predict how well a renormalized fixed decoder will perform on that day. The pilot used the first 46–61 sessions of monkey N (January 2020 to about March 2021) and 283 session pairs at most 60 days apart. Decoders were ridge regressions on 8 lags of spike-band power ($\alpha = 0.1$), trained on the first 300 trials and tested on the rest.

We computed 17 statistics in nine groups that need no labels on the later day:

- the shift in channel means and in channel standard deviations;
- the principal angles between the dominant subspaces of the two days;
- disagreement within a bootstrap ensemble of decoders;

- disagreement between decoders built on different views of the data (spike-band power versus threshold crossings, and 8 lags versus 1 lag);
- statistics of the predicted output (velocity bias, variance ratio, Wasserstein distance to the training kinematics and smoothness);
- self-consistency between predicted position and predicted velocity;
- Procrustes alignment disagreement, the Procrustes residual and the share of decoder readout outside the dominant subspace;
- neural modulation depth;
- the log ratio of impedances and the number of silent channels.

Elapsed time predicted decoder accuracy well (Spearman $\rho = -0.75$ across all pairs and -0.56 within 60 days). After controlling for log elapsed days, the partial Spearman correlation of every statistic with R^2 was at most 0.27 (output variance) and mostly below 0.1. In time-blocked cross-validated prediction of R^2, elapsed days alone gave a mean absolute error of 0.064. The label-free statistics alone gave 0.067–0.071, and elapsed days together with the statistics gave 0.067–0.070. When the target was the gain from recalibration rather than R^2, elapsed days gave 0.045 and the statistics gave 0.054. For detecting unexpectedly poor days (residual more than one standard deviation below the time trend), every statistic had an area under the receiver operating characteristic curve of 0.50–0.58.

The residual variation was not noise. Split-half reliability of the residuals, computed on alternating pairs of trials, was $r = 0.72$, or 0.84 after Spearman–Brown correction. Alternating pairs were needed because center-out trials alternate between outward and return targets, which made odd/even single-trial splits negatively correlated. Day-to-day performance beyond the time trend is therefore real and reliable, but none of the label-free statistics we tried captured it. Abrupt faults on single channels remain detectable by construction, and such faults may still justify monitoring.

Supplementary Note 2: Simulator

The simulator (`ibci.sim`) is a feature-level generative model of chronic change. It transforms a real base session into a simulated session Δt days later, so that behavior and latent dynamics stay realistic. Its components are defined in the Methods and its parameters are listed in Supplementary Table 12. Two further components, heavy-tailed gain jumps and random walks of raw offsets and scales, are included for decoders that are not renormalized. They do not affect renormalized decoders and were not calibrated.

Calibration. The simulated ladder must be scored with exactly the same correction code and regularization as the real ladder. An earlier calibration used regularization 100–1,000 times stronger than the values chosen on real data. That handicapped the simulated corrections, and the optimizer compensated with distorted parameters (loss 0.663 versus 0.074 after the fix). We also restricted silencing to channels that were active in the base session, because most channels are already silent on any given day.

Validation. The simulator calibrated on days before 700 was evaluated on base sessions after day 700, for all rungs of the ladder and for 20, 50 and 100 labeled trials. Two references were used: the simulator with all drift switched off, and the median

retention measured on real pairs before day 700 (an empirical curve). The simulator beat the no-drift reference everywhere. It beat the empirical curve only for corrections it was not fitted to; for fitted corrections and for different amounts of labeled data the empirical curve was as good or better (main text, Fig. 8b). The simulator is therefore most useful for predicting how untested corrections behave.

Intended uses and limits. The simulator is meant for stress-testing decoders and label-free methods under controlled drift and electrode switching, and for comparing recalibration policies. It is calibrated on one monkey with Utah arrays, spike-band power and a finger task, and it has not been calibrated to human recordings. Its mechanisms reproduce the measured statistics, which does not show that they are the true physical mechanisms. Users can calibrate it to other datasets with `scripts/calibrate_sim.py`.

Supplementary Note 3: Analysis choices revised during the study

We report analysis choices that we revised after internal review, because each could mislead similar studies.

One-day pairs included two-day gaps. Session pairs were matched to a target gap within 25% or one day, whichever was larger, so the one-day set contains two-day gaps. In participant T6 most "one-day" pairs were two-day pairs, which changed the apparent overnight retention. We now report one- and two-day pairs separately.

Decoder regularization confounds decay. With the small ridge penalties used in the original dataset papers, decoders decayed faster than with a tuned penalty, especially in monkey N. Decay is therefore reported for decoders whose penalty was tuned.

Label-free alignment was compared with renormalization. In our ladder, alignment was first applied to renormalized features, which cannot separate the two. We therefore also aligned mean-updated features.

Electrode revival needs a null. Counting electrodes that fall silent and later recover suggested a striking recovery rate. The same statistic computed on shuffled session orders was almost as high, so only long silences carry evidence of a temporal process. Electrode-level tests also treat the array, not the electrode, as the unit, because neighboring electrodes are correlated.

Flexible corrections need scrambled-channel controls. Re-weighting one weight per channel recovered most of the loss in the human participants, which at first suggested that drift there is mostly per-channel. Decoders with scrambled channels could also be re-weighted to high accuracy when the decoded variable was two-dimensional, so high recovery alone does not show per-channel drift. We therefore report scrambled-channel and non-negative controls (Supplementary Table 5).

A full input remap is not a structural test. A free $C \times C$ linear map in front of a frozen decoder can reproduce any decoder of the same form when the readout rank is below the number of channels. We used it only as a calibration target for the simulator.

Recurrent networks memorize closed-loop sessions. A 256-unit network trained without early stopping reached training R^2 of 0.97 and test R^2 of -1.1 on a human

session. All network results use a smaller, regularized network with early stopping on held-out training trials.

Small-budget recalibration shrinks the intercept. Refitting the intercept freely with 10–20 trials shifted the output offset and made recalibration harmful in many pairs.

Supplementary Note 4: Excluded participants and sensitivity analyses

Participants whose decoders did not work. The BrainGate decoding data contain nine participants. Retention is defined only when the reference decoder, trained on the test day, works. We therefore required a median reference R^2 above 0.1 across a participant's session pairs (Methods). The rule was fixed when the first added participant (T3) failed and before the results of the other added participants were seen. T3 (8 sessions, 12 pairs) and T2 (42 sessions, 120 pairs) failed it, with median reference R^2 of -0.05 and -0.02. Their movement periods lasted a median of 15.6 s, against 1.6–3.9 s in the included participants (Supplementary Table 14). For T2, block-wise normalization and threshold crossings did not raise same-day R^2 above about 0.1 except on its earliest session, whereas the same checks left T6 and T7 working.

T11. T11 (196 sessions, 677 pairs) also failed the rule (median reference R^2 -16.19; 155 pairs above 0.1). In 107 of 196 sessions, the held-out trials contained spike-band power more than 10^3 standard deviations from the training trials of the same session. This was rare in the included participants (T7 2 of 35, T8 4 of 58, T10 1 of 57 sessions), and such pairs are removed by the pair rule below. The jumps coincided with recording blocks. When each block was z-scored separately, as in the dataset paper (Hahn et al., 2026), most affected T11 sessions decoded normally, and clean sessions were unchanged. Block-wise z-scoring uses the statistics of every block, including blocks in the evaluation segment, so this analysis uses more unlabeled information from the test day than renormalization does. With block-wise normalization, the median reference R^2 of T11 was 0.14 (351 of 677 pairs above 0.1). The renormalized fixed decoder kept a median retention of 0.53 after one day, -0.06 after 30 days and -0.49 after 480 days, and re-weighting each channel restored retention to 0.86–0.99, in line with the included participants. T11 is not part of the main analysis because the inclusion rule was fixed in advance.

Session pairs with a failed reference decoder. Within included individuals, pairs whose reference R^2 was at most 0.1 were left out (0–18% of pairs per individual). With all pairs kept, the main conclusions did not change. Time to half retention ranged from <1 days (T8) to 125 days (T6), compared with <1 to 138 days with the rule, so the spread between individuals remained about a hundredfold. The largest single change was in T7, whose retention after one day lay close to one half (0.59 with all pairs, 0.48 with the rule), so its time to half retention moved from 7.1 days to under one day. T6 changed from 125 to 138 days, which is small relative to its slow decay, and the other individuals changed by less than one day. Full results with all pairs are in `results/v2_summary_allpairs.json`.

Supplementary Note 5: Pretrained foundation decoder

Pretraining data. The public NDT3 checkpoint pretrained on about 2,000 hours (45 million parameters) stores the list of its pretraining sessions: 26,298 sessions from 23 subjects, mostly human sessions from the University of Pittsburgh plus monkey reaching datasets. The monkey datasets include those of monkeys C and M of this study, so we did not test them. No BrainGate session and no session of monkey N is listed.

Choice of features and individuals. NDT3 takes threshold-crossing counts. With threshold crossings, the same-day ridge decoder reached a median R^2 of 0.22 in monkey N (0.29 with spike-band power), 0.50 in T5 (0.54) and 0.32 in T8 (0.34). The crossings of T6 at -4.5 times the RMS were too sparse to decode (0.04, against 0.28 with spike-band power), so T6 was not used. We chose monkey N and the two participants with the best threshold-crossing decoding and the longest recordings among those whose decoders worked well.

Learning rate. A first run used a learning rate of 10^{-4}. In T5, NDT3's same-day R^2 then fell below the ridge decoder in most sessions. On one T5 session (ridge decoder R^2 0.49), a learning rate of 4×10^{-4} gave 0.38, against 0.26 with 10^{-4}; longer training gave 0.30, and a shorter context during evaluation 0.29. Repeated runs with identical settings differed by up to 0.07. We then fixed the rule described in Methods: the rate with the higher NDT3 validation R^2 on each individual's earliest session (monkey N 0.44 with 4×10^{-4} versus 0.42; T5 0.61 versus 0.59; T8, on the array kept by the array rule, 0.37 with 10^{-4} versus 0.34). Monkey N and T5 were rerun completely with the chosen rate. In monkey N, the first run (10^{-4}) gave the same pattern: same-day R^2 0.39 and median retention 1.08 with 300 trials.

A nearly silent array in T8. With both arrays, NDT3 did not learn T8 (R^2 0.00 on its earliest session). Training the same architecture from scratch gave -0.01, and the first array alone -0.02. With the second array alone, NDT3 reached 0.31, against 0.38 for the ridge decoder on all channels. Only 17 of the first array's 96 channels were active, and the ridge decoder reached 0.02 on that array alone and 0.38 with or without it, whereas NDT3 failed whenever it was present. We therefore fixed the array rule in Methods before running T8's pairs. It removed this array only.

Evaluation. Predictions were made in overlapping 2-s windows advancing by 1 s, keeping the last second of each. On a monkey N session this gave R^2 of 0.39 against 0.40 for NDT3's streaming evaluation with 1 s of context, and 0.08 against 0.09 seven days later. NDT3's data loader drops windows whose behaviour is constant, which happens when the cursor-to-target direction does not change for 2 s. Because behaviour inputs are masked during prediction, we added a ramp of 10^{-3} to the behaviour written for evaluation windows so that every window was kept; predictions were unchanged.

Scope. The pilot used one model size, threshold crossings and 12 session pairs in each of three individuals, all offline. A rule fixed before the pilot required NDT3 to match or exceed the ridge decoder on same-day data in at least two of three individuals before a full benchmark run. It was met only in monkey N, so the analysis was not extended. Larger models, other adaptation schemes or more labeled data may behave differently.

Supplementary Table 1 Datasets. Session counts are per analysis; some human sessions contribute to more than one analysis. SBP, spike-band power; TC, threshold crossings.

Individual	Source	Arrays and site	Signal	Task	Sessions (span)
Monkey N	DANDI 001201 (LINK)	2 Utah, motor cortex	SBP, TC	finger flexion	312 on 303 days (3.4 years)
Monkey C	DANDI 000688	Utah, motor/premotor	sorted spikes	reaching	68 (3.0 years)
Monkey M	DANDI 000688	Utah, motor/premotor	sorted spikes	reaching	28 (1.5 years)
Human T6	Dryad x0k6djj1h	1 Utah, motor cortex	SBP	closed-loop cursor	124 (3.1 years)
Human T5	Dryad x0k6djj1h	2 Utah, motor cortex	SBP	closed-loop cursor	92 (7.2 years)
Human T9	Dryad x0k6djj1h	2 Utah, motor cortex	SBP	closed-loop cursor	117 (2.0 years)
Human T7	Dryad x0k6djj1h	2 Utah, motor cortex	SBP	closed-loop cursor	35 (1.4 years)
Human T10	Dryad x0k6djj1h	1 Utah, motor cortex	SBP	closed-loop cursor	57 (0.9 years)
Human T8	Dryad x0k6djj1h	2 Utah, motor cortex	SBP	closed-loop cursor	58 (2.7 years)
Human T2 (excluded)	Dryad x0k6djj1h	1 Utah, motor cortex	SBP	closed-loop cursor	42 (2.0 years)
Human T3 (excluded)	Dryad x0k6djj1h	1 Utah, motor cortex	SBP	closed-loop cursor	8 (1.0 years)
Human T11 (excluded)	Dryad x0k6djj1h	2 Utah, motor cortex	SBP	closed-loop cursor	196 (4.6 years)
14 humans	Dryad x0k6djj1h	20 Utah arrays	TC rates, impedance	research sessions	2,289 ($\leq$7.3 years)
T11, T5	Dryad n2z34tn5s (MINDFUL)	motor cortex	SBP (T11), TC (T5)	closed-loop cursor	15 and 6 days

Supplementary Table 2 Correction ladder by individual and gap (linear decoder, tuned penalty $\alpha = 10^4$). Median decoding R^2 for each rung, and median retention of the renormalized fixed decoder with 95% cluster-bootstrap confidence intervals. Each training session contributes at most one pair per gap. Supervised rungs used up to 300 labeled trials. Aligned, Procrustes rotation after renormalization; stable aligned, the same estimated from stable channels.

Gap (days)	Pairs	Reference	None	Mean update	Renorm.	Aligned	Stable aligned	Re-weighted	Ridge to prior	Retention [95% CI]
Monkey N										
1	30	0.29	−0.61	0.19	0.24	0.21	0.21	0.23	0.31	0.79 [0.74, 0.85]
7	48	0.30	−0.75	0.16	0.19	0.16	0.16	0.20	0.30	0.63 [0.60, 0.68]
30	61	0.28	−1.46	0.07	0.11	0.11	0.08	0.12	0.29	0.41 [0.34, 0.57]
120	70	0.29	−1.48	0.06	0.08	0.06	0.05	0.09	0.29	0.30 [0.25, 0.35]
480	50	0.28	−0.95	0.03	0.00	−0.01	−0.00	0.08	0.29	0.01 [−0.03, 0.09]
Monkey C										
1	37	0.27	0.10	0.20	0.20	0.19	−0.10	0.24	0.28	0.73 [0.65, 0.80]
7	31	0.33	0.02	0.20	0.22	0.19	−0.06	0.27	0.33	0.62 [0.55, 0.70]
30	11	0.45	−0.15	0.10	0.10	0.11	0.09	0.15	0.45	0.23 [0.19, 0.55]
120	18	0.18	−0.61	−0.00	0.01	−0.03	−0.21	0.07	0.18	0.04 [−0.16, 0.17]
480	33	0.15	−0.78	−0.15	−0.06	−0.29	−0.36	0.02	0.14	−0.12 [−0.42, −0.10]
Monkey M										
1	13	0.36	0.17	0.27	0.27	0.25	−0.02	0.31	0.37	0.80 [0.74, 0.86]
7	6	0.27	0.06	0.15	0.18	0.13	−0.11	0.19	0.27	0.62 [0.51, 0.70]
30	6	0.43	−0.27	0.22	0.26	0.27	0.24	0.30	0.44	0.57 [0.48, 0.68]
120	9	0.25	−0.23	0.00	0.02	0.01	−0.22	0.07	0.22	0.09 [−0.07, 0.19]
480	9	0.35	−0.23	−0.03	0.01	−0.07	−0.15	0.07	0.35	0.04 [−0.47, 0.11]
Human T6										
1	23	0.31	−0.58	0.17	0.18	0.20	0.20	0.27	0.30	0.79 [0.55, 0.84]
7	57	0.25	−0.36	0.10	0.15	0.15	0.14	0.20	0.22	0.74 [0.63, 0.84]
30	79	0.26	−0.52	0.08	0.20	0.21	0.20	0.26	0.28	0.76 [0.70, 0.83]
120	98	0.26	−0.74	0.04	0.16	0.14	0.13	0.23	0.26	0.60 [0.48, 0.66]
480	76	0.30	−1.37	0.03	0.12	0.12	0.12	0.26	0.29	0.35 [0.26, 0.44]
Human T5										
1	17	0.56	−1.02	0.12	0.42	0.47	0.48	0.53	0.51	0.73 [0.34, 0.83]
7	25	0.55	−1.70	−0.37	0.26	0.26	0.28	0.45	0.51	0.52 [0.23, 0.68]
30	49	0.48	−3.88	−0.85	0.13	0.10	0.12	0.39	0.46	0.28 [−0.02, 0.56]
120	72	0.53	−3.73	−1.13	0.07	−0.01	−0.08	0.42	0.50	0.13 [−0.02, 0.21]
480	84	0.52	−6.10	−0.96	−0.09	−0.15	−0.13	0.42	0.49	−0.17 [−0.28, −0.06]
Human T9										
1	23	0.32	−0.92	−0.01	0.18	0.14	0.14	0.27	0.31	0.60 [0.40, 0.68]
7	53	0.33	−0.97	−0.10	0.11	0.10	0.10	0.28	0.31	0.40 [0.32, 0.51]
30	103	0.35	−4.08	−0.37	0.10	0.07	0.10	0.27	0.33	0.31 [0.15, 0.43]
120	90	0.36	−4.46	−0.31	0.04	−0.02	0.00	0.25	0.33	0.14 [0.01, 0.19]
480	51	0.36	−7.40	−0.45	−0.15	−0.15	−0.18	0.25	0.34	−0.38 [−0.53, −0.35]
Human T7										
1	4	0.35	−1.59	−0.88	0.20	0.14	0.11	0.28	0.35	0.53 [0.20, 0.72]
7	8	0.28	−2.69	−0.30	0.15	0.15	0.14	0.23	0.26	0.51 [−0.09, 0.71]
30	19	0.31	−2.77	−1.22	0.09	0.09	0.06	0.25	0.31	0.29 [−0.02, 0.46]
120	30	0.34	−6.88	−1.24	−0.02	−0.05	−0.09	0.26	0.32	−0.08 [−0.54, 0.13]
480	12	0.57	−2.46	−0.74	0.03	−0.10	−0.06	0.44	0.57	0.06 [−0.20, 0.09]
Human T10										
1	16	0.34	−0.06	−0.01	0.20	0.20	0.17	0.24	0.31	0.58 [0.04, 0.79]
7	26	0.30	−1.35	−0.14	0.12	0.11	0.12	0.22	0.28	0.42 [0.27, 0.65]

Supplementary Table 3 Label-free corrections. Median over training sessions of the paired difference in R^2 (each training session's pairs at all gaps, 1 to 480 days; pairs with a working reference decoder), with two-sided Wilcoxon signed-rank P in parentheses. Mean, mean updating only; rot., Procrustes rotation; stab., factor-analysis stabilizer; FA, factor analysis.

Individual	Renorm. vs mean	Mean+rot. vs mean	Renorm.+rot. vs renorm.	CORAL vs renorm.	Stab. vs renorm.	Stab. vs fixed FA
Monkey N	0.02 (1e-10)	−0.06 (4e-11)	−0.02 (4e-11)	−0.00 (0.95)	−0.06 (9e-14)	−0.02 (2e-11)
Monkey C	0.02 (2e-4)	−0.07 (6e-10)	−0.02 (7e-8)	−0.00 (9e-5)	−0.11 (8e-6)	−0.06 (9e-9)
Monkey M	0.01 (5e-5)	−0.08 (1e-6)	−0.03 (3e-4)	−0.00 (0.77)	−0.15 (5e-7)	−0.09 (5e-7)
Human T6	0.09 (2e-19)	−0.07 (6e-11)	−0.01 (5e-5)	−0.01 (2e-8)	−0.08 (4e-16)	−0.06 (9e-17)
Human T5	2.10 (3e-16)	−9.19 (2e-15)	−0.05 (1e-5)	−0.08 (1e-15)	0.02 (0.038)	0.02 (0.14)
Human T9	0.44 (8e-20)	−2.56 (1e-18)	−0.05 (1e-12)	−0.02 (3e-12)	−0.11 (2e-11)	−0.03 (2e-7)
Human T7	1.34 (5e-10)	−8.69 (5e-10)	−0.03 (0.011)	−0.03 (5e-6)	−0.03 (0.17)	0.02 (0.2)
Human T10	0.88 (6e-10)	−4.40 (5e-10)	−0.00 (0.25)	−0.02 (1e-8)	−0.03 (0.0028)	−0.02 (0.014)
Human T8	1.74 (1e-10)	−2.24 (4e-6)	−0.06 (4e-9)	−0.01 (3e-4)	−0.09 (0.014)	0.00 (0.93)

a

Supplementary Table 4 Monitoring. a, Offline, monkey N (537 pairs): Spearman correlation with decoder R^2, raw and partial given log elapsed days, and time-blocked cross-validated error of predicted R^2 (days and divergence uses both divergences). **b**, Closed loop, public MINDFUL data: Pearson correlation of angular error with elapsed days and each divergence (raw and partial given days), and leave-one-day-out mean absolute error (degrees).

Predictor	Raw ρ	Partial ρ
Elapsed days (log)	−0.67	–
Neural divergence	−0.34	−0.16
Output divergence	−0.22	−0.16
Sum	−0.36	−0.17
Cross-validated error of predicted R^2 (mean absolute error)		
Days only	0.085	
Divergence only	0.114	
Days and divergence	0.090	

b

				Neural		Output		Error (°)			
	Days	Windows	r(days)	raw	partial	raw	partial	days	+neural	+output	output
T11	15	145	0.84	0.82	0.37	0.81	0.73	17.1	17.5	11.9	12.8
T5	6	73	0.17	0.50	0.49	0.79	0.78	32.9	38.3	20.8	15.0

Supplementary Table 5 Channel controls and electrode link. For 40 training sessions per individual (22 for monkey M and 33 for T7; tuned penalty), median retention of the renormalized fixed decoder, after re-weighting the correctly assigned decoder, after re-weighting a decoder with scrambled channels, and the same two with non-negative weights. Electrode link (humans): median over training sessions of the Spearman correlation between fitted channel weights and the change in each electrode's log threshold-crossing rate or log impedance between the two days, with the share of sessions with a positive correlation.

			Signed weights		Non-negative weights		Electrode link	
Individual	Pairs	No labels	Correct	Scrambled	Correct	Scrambled	Firing rate	Impedance
Monkey N	134	0.39	0.46	−0.19	0.49	−0.17	–	–
Monkey C	84	0.42	0.62	0.02	0.59	0.01	–	–
Monkey M	43	0.49	0.68	0.01	0.67	−0.00	–	–
Human T6	113	0.61	0.89	0.65	0.89	0.36	0.05 (67%)	−0.02 (46%)
Human T5	112	0.13	0.83	0.77	0.72	0.57	0.03 (77%)	−0.01 (33%)
Human T9	108	0.15	0.78	0.56	0.72	0.29	0.11 (92%)	−0.07 (14%)
Human T7	73	0.09	0.80	0.63	0.67	0.38	0.04 (68%)	−0.06 (21%)
Human T10	92	0.23	0.72	0.57	0.65	0.35	0.06 (87%)	−0.03 (36%)
Human T8	117	−0.11	0.52	0.33	0.38	0.10	0.07 (90%)	−0.06 (18%)

Supplementary Table 6 Recurrent network versus linear decoder. For 25 training sessions per individual: median reference R^2, retention of the renormalized fixed decoder, and share of the loss recovered by re-weighting each channel.

		Reference R^2		Retention		Loss recovered	
Gap (days)	Pairs	Linear	Network	Linear	Network	Linear	Network
Monkey N							
1	8	0.28	0.40	0.77	0.77	−0.07	0.63
7	14	0.30	0.40	0.67	0.73	−0.24	0.52
30	19	0.28	0.36	0.41	0.48	−0.04	0.40
120	23	0.29	0.40	0.19	0.19	0.13	0.48
480	17	0.28	0.38	−0.00	−0.07	0.12	0.36
Monkey C							
1	17	0.31	0.66	0.71	0.81	0.41	0.68
7	10	0.35	0.74	0.59	0.73	0.38	0.52
30	4	0.50	0.87	0.23	0.15	0.35	0.45
120	6	0.25	0.71	0.00	0.04	0.41	0.57
480	12	0.14	0.32	−0.13	−0.39	0.22	0.45
Monkey M							
1	13	0.36	0.76	0.80	0.70	0.28	0.69
7	6	0.27	0.73	0.62	0.53	0.22	0.62
30	6	0.43	0.68	0.57	0.48	0.34	0.69
120	9	0.25	0.58	0.09	−0.50	0.35	0.64
480	9	0.35	0.83	0.04	−0.33	0.19	0.48
Human T6							
1	4	0.27	0.19	0.68	0.08	0.33	0.85
7	12	0.24	0.22	0.65	0.64	0.72	1.01
30	15	0.25	0.25	0.74	0.55	0.66	0.42
120	18	0.26	0.27	0.58	0.51	0.65	0.92
480	16	0.32	0.29	0.40	0.49	0.75	0.92
Human T5							
1	3	0.55	0.52	0.83	0.90	0.77	0.39
7	8	0.55	0.52	0.49	0.22	0.83	0.69
30	12	0.50	0.45	0.25	0.06	0.68	0.76
120	20	0.53	0.49	−0.17	−0.62	0.72	0.71
480	23	0.53	0.49	−0.27	−0.53	0.82	0.85
Human T9							
1	3	0.35	0.26	−0.43	−0.46	0.68	0.83
7	10	0.27	0.24	0.36	0.07	0.84	0.83
30	15	0.38	0.33	0.46	0.24	0.73	0.95
120	15	0.39	0.25	−0.02	−0.28	0.73	0.99
480	10	0.36	0.35	−0.33	−1.08	0.74	0.82
Human T7							
1	3	0.36	0.45	0.66	0.35	0.65	0.46
7	5	0.34	0.31	0.56	0.56	0.83	0.90
30	10	0.35	0.29	0.41	0.44	0.81	0.96
120	17	0.40	0.39	0.05	−0.04	0.80	0.82
480	10	0.57	0.57	−0.00	−0.34	0.80	0.84
Human T10							
1	7	0.35	0.28	0.83	0.48	0.72	0.83
7	12	0.31	0.26	0.37	−0.09	0.64	0.87
30	13	0.31	0.28	0.33	−0.34	0.71	0.81
120	10	0.27	0.26	−0.49	−1.39	0.58	0.40
Human T8							
1	2	0.36	0.33	0.28	−0.12	0.15	0.62
7	10	0.32	0.26	0.07	−1.07	0.08	0.27
30	12	0.40	0.37	0.22	−0.31	0.45	0.63
120	19	0.27	0.28	−0.21	−1.01	0.44	0.38
480	16	0.50	0.41	−0.23	−0.73	0.75	0.81

Supplementary Table 7 Matched-output control in the reaching monkeys (tuned penalty). Median retention of the renormalized fixed decoder and after re-weighting, and the share of the loss recovered by re-weighting (%), when decoding hand kinematics (four outputs) or hand-movement direction only (a two-dimensional unit vector, as for the human participants).

Individual	Target	Gap (days)	Renormalized	Re-weighted	Recovered (%)
Monkey C	kinematics	1	0.73	0.81	37
		7	0.62	0.73	38
		30	0.23	0.48	30
		120	0.04	0.35	36
		480	−0.12	0.10	22
	direction	1	0.83	0.90	48
		7	0.74	0.85	46
		30	0.37	0.71	60
		120	0.07	0.46	45
		480	0.03	0.38	36
Monkey M	kinematics	1	0.80	0.83	28
		7	0.62	0.70	22
		30	0.57	0.71	34
		120	0.09	0.34	35
		480	0.04	0.21	19
	direction	1	0.81	0.89	48
		7	0.73	0.87	36
		30	0.66	0.84	47
		120	0.34	0.52	30
		480	0.19	0.56	35

Supplementary Table 8 Angles between decoders fitted on different days (degrees, medians over pairs). Global, angle that allows one global scale; per channel, angle after fitting one scale per channel; permuted, the per-channel angle after randomly permuting the channels of the earlier decoder.

Individual	Gap (days)	Global	Per channel	Permuted
Monkey N	1	44	29	47
	7	50	33	49
	30	60	37	49
	120	65	40	48
	480	74	42	49
Monkey C	1	43	35	80
	7	52	40	75
	30	72	58	70
	120	75	56	78
	480	85	74	79
Monkey M	1	41	31	73
	7	49	39	79
	30	61	47	72
	120	72	53	70
	480	81	64	73
Human T6	1	47	31	53
	7	51	34	52
	30	53	35	53
	120	59	39	52
	480	64	42	52
Human T5	1	57	38	54
	7	65	43	53
	30	71	45	53
	120	79	48	54
	480	84	50	53
Human T9	1	52	33	49
	7	55	35	49
	30	59	37	49
	120	68	41	49
	480	77	44	49
Human T7	1	59	35	49
	7	59	36	52
	30	66	40	49
	120	72	43	48
	480	82	47	48
Human T10	1	49	34	50
	7	57	36	49
	30	61	39	49
	120	69	43	48
Human T8	1	47	28	47
	7	56	32	48
	30	62	36	49
	120	69	39	48
	480	82	45	48

Supplementary Table 9 Regularization by the 1% rule. The rule chose the largest ridge penalty whose median same-day R^2 was within 1% of the best. Columns give median R^2 with $\alpha = 0.1$ and with the chosen α, the median paired gain with 95% cluster-bootstrap confidence interval, and the share of training sessions whose mean gain was positive.

Individual	Chosen α	Gap (days)	Pairs	R^2, $\alpha = 0.1$	R^2, rule	Gain [95% CI]	Share positive
Monkey N	10^4	same day	40	0.30	0.30	0.003 [0.001, 0.006]	0.72
Monkey N	10^4	1	11	0.27	0.28	0.021 [0.008, 0.035]	1.00
Monkey N	10^4	7	24	0.14	0.18	0.035 [0.026, 0.041]	0.92
Monkey N	10^4	30	28	0.10	0.14	0.042 [0.038, 0.052]	1.00
Monkey N	10^4	120	35	0.04	0.10	0.056 [0.050, 0.083]	0.97
Monkey N	10^4	480	28	−0.10	−0.00	0.090 [0.082, 0.112]	1.00
Monkey C	10^4	same day	40	0.21	0.22	0.002 [0.001, 0.003]	0.72
Monkey C	10^4	1	26	0.16	0.18	0.008 [0.005, 0.017]	0.88
Monkey C	10^4	7	22	0.20	0.21	0.012 [0.006, 0.019]	0.77
Monkey C	10^4	30	9	0.10	0.10	0.001 [−0.002, 0.011]	0.67
Monkey C	10^4	120	12	−0.01	0.01	0.031 [0.016, 0.045]	1.00
Monkey C	10^4	480	25	−0.10	−0.07	0.032 [0.014, 0.036]	1.00
Monkey M	10^3	same day	22	0.35	0.35	0.000 [0.000, 0.001]	0.73
Monkey M	10^3	1	14	0.26	0.26	0.001 [0.000, 0.001]	0.79
Monkey M	10^3	7	6	0.16	0.17	0.000 [−0.000, 0.002]	0.67
Monkey M	10^3	30	6	0.25	0.25	0.001 [0.000, 0.012]	1.00
Monkey M	10^3	120	9	0.01	0.01	0.001 [0.000, 0.002]	1.00
Monkey M	10^3	480	9	0.01	0.01	0.001 [0.000, 0.001]	1.00
Human T6	10^4	same day	40	0.22	0.24	0.008 [0.003, 0.013]	0.95
Human T6	10^4	1	6	0.13	0.17	0.011 [0.002, 0.075]	1.00
Human T6	10^4	7	23	0.10	0.14	0.025 [0.016, 0.032]	1.00
Human T6	10^4	30	32	0.16	0.17	0.016 [0.013, 0.030]	0.97
Human T6	10^4	120	38	0.13	0.17	0.018 [0.009, 0.030]	0.95
Human T6	10^4	480	27	0.11	0.12	0.017 [0.005, 0.027]	0.89
Human T5	10^4	same day	40	0.51	0.52	0.008 [0.004, 0.014]	0.97
Human T5	10^4	1	13	0.37	0.45	0.038 [0.013, 0.073]	1.00
Human T5	10^4	7	13	0.26	0.26	0.018 [0.005, 0.053]	1.00
Human T5	10^4	30	25	−0.08	0.10	0.040 [0.012, 0.069]	1.00
Human T5	10^4	120	33	−0.02	0.01	0.040 [0.018, 0.053]	0.97
Human T5	10^4	480	40	−0.10	−0.03	0.029 [0.015, 0.060]	0.97
Human T9	3×10^4	same day	40	0.30	0.31	0.007 [0.000, 0.011]	0.65
Human T9	3×10^4	1	12	0.16	0.18	0.029 [0.013, 0.058]	0.92
Human T9	3×10^4	7	22	0.06	0.12	0.031 [0.012, 0.077]	0.82
Human T9	3×10^4	30	39	0.08	0.11	0.037 [0.027, 0.051]	0.87
Human T9	3×10^4	120	40	0.02	0.07	0.041 [0.029, 0.064]	0.90
Human T9	3×10^4	480	20	−0.17	−0.11	0.043 [0.035, 0.077]	1.00
Human T7	3×10^4	same day	33	0.24	0.26	0.007 [0.004, 0.024]	0.79

a

Supplementary Table 10 Recalibration with few labeled trials. a, Monkey N (86 pairs): median retention without labels and, for each number of labeled trials, ridge to prior / fresh decoder / re-weighting (regularization by cross-validation). **b**, Human participants (30 training sessions each, all gaps): re-weighting / ridge to prior / fresh decoder. **c**, Harmful recalibrations in monkey N (more than 0.01 R^2 below the renormalized fixed decoder) with the shrinkage chosen by cross-validation or fixed at the value best on other training sessions, with P from a Wilcoxon signed-rank test over training sessions, and median retention for both and for the best value in hindsight.

Gap (days)	Pairs	No labels	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$
1	7	0.64	0.47 / −0.18 / 0.02	0.63 / 0.22 / 0.50	0.75 / 0.55 / 0.64	0.83 / 0.70 / 0.69	1.03 / 1.02 / 0.73
7	18	0.48	0.51 / 0.17 / 0.38	0.58 / 0.39 / 0.43	0.77 / 0.68 / 0.54	0.89 / 0.81 / 0.61	1.02 / 1.01 / 0.54
30	20	0.15	0.21 / −0.07 / 0.05	0.27 / 0.25 / 0.09	0.60 / 0.61 / 0.25	0.79 / 0.80 / 0.29	1.02 / 1.02 / 0.27
120	26	0.13	0.31 / 0.24 / −0.02	0.44 / 0.43 / 0.06	0.67 / 0.66 / 0.24	0.82 / 0.83 / 0.19	1.01 / 1.01 / 0.30
480	15	−0.21	0.19 / 0.19 / −0.27	0.35 / 0.42 / −0.07	0.60 / 0.64 / 0.16	0.82 / 0.83 / 0.09	1.01 / 1.01 / 0.27

b

Participant	Pairs	No labels	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$
Human T6	41	0.56	0.54 / 0.63 / 0.36	0.53 / 0.68 / 0.47	0.64 / 0.72 / 0.68	0.61 / 0.75 / 0.76	0.84 / 0.90 / 0.92
Human T5	57	0.21	0.22 / 0.53 / 0.35	0.43 / 0.60 / 0.57	0.58 / 0.70 / 0.72	0.67 / 0.76 / 0.80	0.81 / 0.89 / 0.91
Human T9	24	0.14	0.26 / 0.14 / 0.12	0.27 / 0.27 / 0.55	0.46 / 0.36 / 0.79	0.65 / 0.50 / 0.81	0.75 / 0.93 / 0.98
Human T7	24	0.09	0.48 / 0.54 / 0.52	0.63 / 0.65 / 0.83	0.70 / 0.79 / 0.93	0.69 / 0.82 / 0.92	0.78 / 0.99 / 0.99
Human T10	40	0.29	0.22 / 0.25 / 0.11	0.23 / 0.36 / 0.34	0.44 / 0.49 / 0.60	0.59 / 0.60 / 0.74	0.72 / 0.89 / 0.94
Human T8	59	0.04	−0.07 / 0.21 / 0.10	0.06 / 0.32 / 0.16	0.18 / 0.36 / 0.41	0.29 / 0.44 / 0.46	0.54 / 0.74 / 0.71

c

		Harmful (%)			Median retention		
Trials	Pairs	Cross-validation	Fixed	P (Fisher)	Cross-validation	Fixed	Hindsight
10	86	10.5	1.2	0.027	0.31	0.37	0.40
20	86	8.1	0.0	0.042	0.43	0.44	0.48
50	86	1.2	0.0	0.317	0.64	0.63	0.68
100	86	0.0	1.2	0.317	0.82	0.81	0.83
300	86	0.0	0.0	1.0	1.02	1.01	1.02

a

Supplementary Table 11 Electrode statistics. a, Silence and recovery in initially active human electrodes under different definitions, observed and after shuffling each electrode's session order (shuffled silent episodes in parentheses). **b**, Per array: yield is the percentage of electrodes with a threshold-crossing rate of at least 2 Hz (median of the first / last three sessions); h_{off} and h_{on} are switching rates per 1,000 days; Rev./sil., recovered and silenced initially active electrodes; Kurt., excess kurtosis of session-to-session changes in log rate after removing session-wide changes; Imp. ρ, Spearman correlation of the array median impedance with day; Edge − int., median edge minus median interior slope of log rate per year, with the one-sided Mann–Whitney P (electrodes treated as independent; see main text for array-level tests).

Definition	Silent episodes	Recovered	Recovered (%)	Shuffled (%)
At least 3 sessions below 2 Hz (main definition)	730	584	80	79 (594)
Threshold −3.5 times noise	8	7	88	–
Threshold −5.5 times noise	646	475	74	–
Fixed threshold in microvolts per electrode	840	601	72	72 (654)
Array-wide shifts removed	747	575	77	–
Mean waveform trough at least 30 μV	736	523	71	–
Silence below 0.5 Hz	607	430	71	71 (418)
Silence lasting at least 30 days	168	96	57	41 (81)

b

Part.	Array	Sess.	Span (d)	Yield (%)	h_{off}	h_{on}	Rev./sil.	Kurt.	Imp. ρ	Edge − int.	P
A1	single	48	252	81 / 55	16.3	27.1	44/68	2.9	−0.27	0.69	0.322
S1	single	40	337	3 / 1	46.6	10.7	1/4	0.7	–	−0.14	0.017
S2	single	32	458	1 / 6	24.2	10.9	0/1	–	–	0.38	0.925
S3	single	207	1,954	84 / 10	14.9	3.3	68/88	3.9	−0.91	−0.10	0.040
T1	single	41	277	65 / 86	2.6	27.1	11/17	7.8	−0.93	−0.18	0.379
T10	MFG	104	329	96 / 86	8.6	54.1	56/59	9.8	−0.70	0.27	0.877
T10	dPCG	104	329	21 / 44	7.7	9.1	26/31	3.2	−0.21	−0.84	0.006
T11	lateral	292	1,677	20 / 10	7.0	1.5	16/24	5.5	−0.95	−0.08	0.025
T11	medial	292	1,677	38 / 64	3.7	6.3	35/35	4.1	−0.99	−0.12	0.039
T2	single	176	914	57 / 14	22.4	3.4	50/67	4.1	−0.98	0.50	0.966
T3	single	87	399	0 / 2	34.3	4.0	0/0	–	−0.98	−0.02	0.131
T5	lateral	438	2,665	60 / 30	2.4	1.2	41/59	4.4	−0.94	−0.05	0.055
T5	medial	438	2,665	51 / 40	2.9	2.7	54/59	2.4	−0.90	−0.03	0.271
T6	single	190	1,184	16 / 2	11.0	3.5	12/22	2.7	−0.97	−0.11	0.059
T7	lateral	84	521	24 / 31	20.1	21.7	23/28	1.9	−0.83	−0.05	0.161
T7	medial	84	521	36 / 49	8.7	11.8	34/42	1.5	−0.91	−0.81	<0.001
T8	lateral	329	1,061	11 / 38	1.1	1.2	12/12	6.7	−0.92	0.29	0.986
T8	medial	329	1,061	55 / 41	5.2	2.9	64/70	3.0	−0.92	0.27	0.805
T9	lateral	221	731	14 / 16	18.7	7.5	13/17	4.3	0.26	−0.98	<0.001
T9	medial	221	731	21 / 44	13.1	6.6	24/27	2.1	−0.96	−0.22	0.028
Monkey N	both	303	1,242	38 / 19	3.1	0.8	23/31	10.2	−0.88	−0.15 / −0.09	<0.001 / 0.046

Supplementary Table 12 Simulator parameters. Values calibrated on monkey N sessions before day 700 (early) and from day 700 onwards (late). Switching rates were measured from electrode activity in each period. Fixed parameters: local mixing share $\beta = 0.5$, turnover concentration 5, 20 latent dimensions and silent-signal fraction $\kappa = 0.37$.

Parameter	Unit	Early	Late
Session-to-session mixing s.d.	1	0.959	0.768
Slow mixing s.d. (asymptote)	1	1.478	1.317
Slow mixing time constant	days	15	12
Turnover fraction, next session	1	0.042	0.032
Turnover fraction, asymptote	1	0.120	0.127
Turnover time constant	days	360	527
Active to silent rate	per day	0.0038	0.0034
Silent to active rate	per day	0.0018	0.0017
Calibration loss		0.074	0.063

Supplementary Table 13 Training on simulated futures. Median R^2 on real later sessions of monkey N for ridge decoders trained on sessions after day 700 (simulator calibrated before day 700). Base, $\alpha = 0.1$; stronger ridge, $\alpha = 10^4$; perturbation, random channel dropout and gain noise; simulator, eight simulated futures; simulator and ridge, both; several days, data from the three previous sessions; reference, decoder trained on the test day.

Gap (days)	Pairs	Base	Stronger ridge	Perturbation	Simulator	Simulator and ridge	Several days	Reference
1	9	0.21	0.22	0.20	0.20	0.20	0.23	0.29
7	20	0.18	0.21	0.18	0.19	0.20	0.23	0.34
30	18	0.08	0.13	0.08	0.09	0.10	0.15	0.28
120	18	0.07	0.12	0.07	0.07	0.08	0.14	0.30
240	12	0.05	0.10	0.06	0.08	0.08	0.11	0.29

Supplementary Table 14 BrainGate decoding participants and the inclusion rule. Arrays (channels), sessions and s the closed-loop cursor sessions; median trials (movement periods) per session and median movement duration; median R^2 of th reference decoder across session pairs (linear decoder, default penalty). Participants entered the decay and correction analyses this median exceeded 0.1 (Methods). For T2, neither block-wise normalization nor threshold crossings raised same-day R^2 abo about 0.1 except on its earliest session.

Participant	Arrays (channels)	Sessions	Span (years)	Trials per session	Movement (s)	Reference R^2	In
T6	1 (96)	124	3.1	368	1.6	0.25	
T5	2 (192)	92	7.2	552	1.7	0.50	
T9	2 (192)	117	2.0	275	3.9	0.33	
T7	2 (192)	35	1.4	290	3.8	0.32	
T10	1 (96)	57	0.9	403	3.2	0.26	
T8	2 (192)	58	2.7	722	2.8	0.34	
T2	1 (96)	42	2.0	184	15.6	−0.02	
T3	1 (96)	8	1.0	158	15.6	−0.05	
T11	2 (192)	196	4.6	215	3.3	−16.19	

Supplementary Table 15 Benchmark baselines. Valid pairs per individual and median retention of three reference methods after 1, 30 and 480 days: the renormalized fixed decoder (no labels), re-weighting each channel and a refit shrunk toward the old decoder (both with 300 labeled trials). Linear decoder, tuned penalty. T10 had no 480-day pairs. Per-pair scores and bootstrap intervals are released with the benchmark.

		Renormalized			Re-weighted (300)			Refit to prior (300)		
Individual	Pairs	1 d	30 d	480 d	1 d	30 d	480 d	1 d	30 d	480 d
Monkey N	259	0.79	0.41	0.01	0.80	0.45	0.28	1.02	1.00	1.01
Monkey C	130	0.73	0.23	−0.12	0.81	0.48	0.10	1.00	1.00	1.00
Monkey M	43	0.80	0.57	0.04	0.83	0.71	0.21	1.01	1.01	1.02
Human T6	333	0.79	0.76	0.35	0.88	0.92	0.82	0.92	0.98	0.96
Human T5	247	0.73	0.28	−0.17	0.92	0.82	0.82	0.94	0.94	0.95
Human T9	320	0.60	0.31	−0.38	0.84	0.82	0.70	0.97	0.97	0.97
Human T7	73	0.53	0.29	0.06	0.72	0.84	0.78	0.99	0.99	0.99
Human T10	124	0.58	0.18	–	0.85	0.73	–	0.96	0.95	–
Human T8	155	0.49	−0.03	−0.24	0.66	0.55	0.58	0.60	0.65	0.86

Supplementary Table 16 Pretrained foundation decoder (NDT3) against the ridge decoder. Both decoders received threshold crossings on the same 12 session pairs per individual (gaps of 1 to 480 days). Each cell gives the median R^2 of NDT3 / the ridge decoder, and in parentheses the number of sessions (same day) or pairs where NDT3 was higher. No update: NDT3 applied unchanged versus the renormalized ridge decoder. 50 and 300 trials: NDT3 fine-tuned with the first labeled trials of the new day versus the ridge decoder refit toward its previous weights. Self-supervised: median retention of NDT3 after continued training with its neural prediction loss only. T8's nearly silent first array was left out of NDT3's input by a rule fixed in advance (Methods); the ridge decoder used all channels.

Individual	Pairs	Same day	No update	50 trials	300 trials	Self-supervised
Monkey N	12	0.42 / 0.20 (12/12)	−0.06 / −0.04 (2/12)	0.16 / 0.08 (10/12)	0.41 / 0.18 (12/12)	−0.43
Human T5	12	0.34 / 0.45 (0/11)	−0.24 / −0.19 (3/12)	0.06 / 0.30 (2/12)	0.26 / 0.41 (1/12)	−0.16
Human T8	12	0.22 / 0.28 (4/12)	−0.09 / −0.10 (4/12)	−0.12 / 0.07 (0/12)	−0.04 / 0.18 (1/12)	−0.08